Simulating non-trivial incompressible flows with a quantum lattice Boltzmann algorithm

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AIRBUS

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WHO ARE WE?

PsiQuantum

- A full-stack quantum company
- Employing 450+ experts
- With global presence in US, Europe and Australia

Mission: To build and deploy the first useful quantum computers

This means:

- Fault-tolerance: full error correction
- Large-scale: 1000000+ qubits

AGENDA

01 Motivation

CFD bottlenecks and beyond

02 Framework

Quantum algorithms for CFD

03 Results

Performance of quantum LBE algorithm

04 Conclusions & outlook

Main takeaways & areas for improvement

AGENDA

01 **Motivation**

CFD bottlenecks and beyond

- Classical CFD complexity
- Quantum CFD landscape

02 Framework

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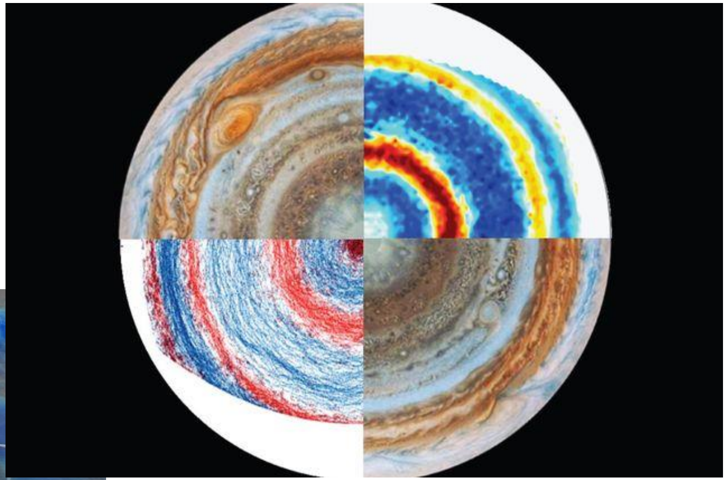
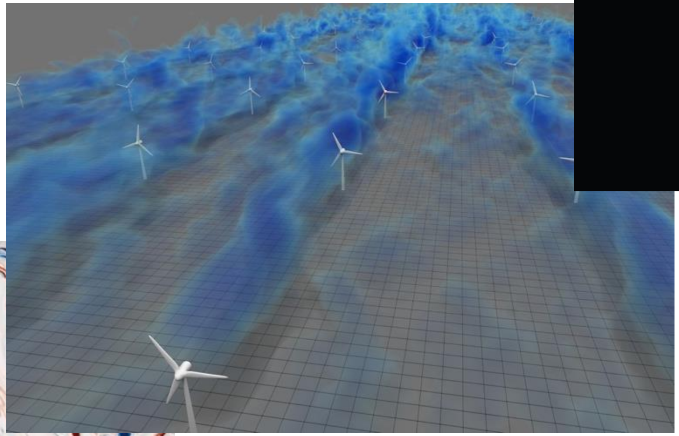
Classical CFD complexity

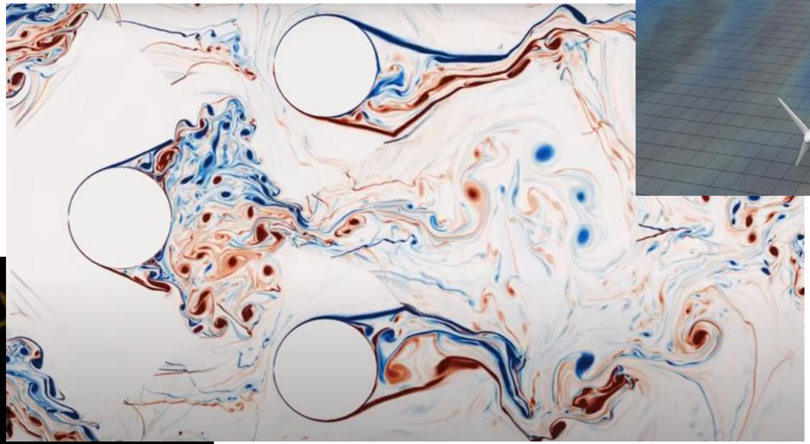
$$\text{Complexity}_C = O(\text{Re}^3)$$



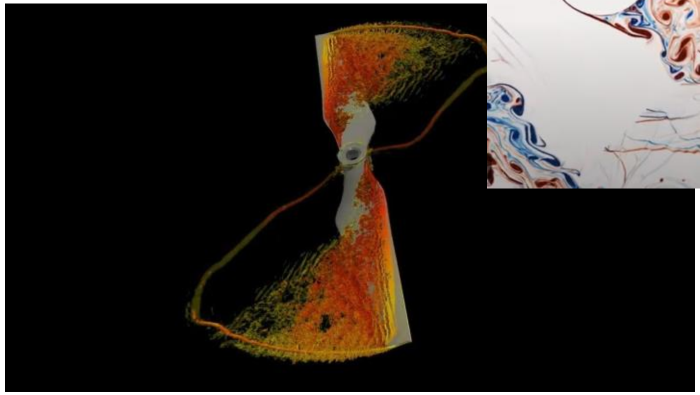
Experiment – Jupiter's jets $\text{Re} \sim 10^{10}$ (UCLA/IRPHE)



LES - wind farm $\text{Re} = 10^8$ (John Hopkins)



DNS - vortex shedding $\text{Re} = 2 \times 10^5$ (John Hopkins)



DNS - starting drone rotor $\text{Re} = 1.5 \times 10^4$ (FAU, KTH, KAUST)

MOTIVATION

Quantum CFD landscape

Paper	Year	Analysis	Speed-up	Resource estimates	Accuracy	CFD apps
Potential quantum advantage for simulation of fluid dynamics Xiangyu Li,[1,*] Xiaolong Yin,[2] Nathan Wiebe,[3,1,4] Jaehun Chun,[1,5] Gregory K. Schenter,[1] Margaret S. Cheung,[1,6] and Johannes Mülmenstädt[1,†] [1]*Pacific Northwest National Laboratory, Richland, WA, USA* [2]*Petroleum Engineering, Colorado School of Mines, Golden, CO, USA* [3]*Department of Computer Science, University of Toronto, Toronto, ON, Canada* [4]*Canadian Institute for Advanced Research, Toronto, ON, Canada* [5]*Levich Institute and Department of Chemical Engineering, CUNY City College of New York, New York, NY, USA* [6]*Department of Physics, University of Washington, Seattle, WA, USA*	2023	Initial	None (claimed exp)	Preliminary	Unverified	No
Detailed assessment of calculating drag force with quantum computers: Explicit time-evolution precludes exponential advantage for nonlinear differential equations John Penuel[1], Amara Katabarwa[2,3], Peter D. Johnson[3], Parker Kuklinski[4], Benjamin Rempfer[4], Collin Farquhar[3], Yudong Cao[*3,6], and Michael C. Garrett[†1] [1]L3Harris Technologies, Inc., Palm Bay, FL, USA [2]Georgia Tech Research Institute, Atlanta, GA, USA [3]Zapata AI, Boston, MA, USA [4]MIT Lincoln Laboratory, Lexington, MA, USA [6]Boston Consulting Group, 200 Pier 4 Blvd, Boston, MA	2024	Flawed end-to-end	Modest poly	Partial	Unverified	No
An end-to-end quantum algorithm for nonlinear fluid dynamics with bounded quantum advantage David Jennings[1], Kamil Korzekwa[*1], Matteo Lostaglio[1], Richard Ashworth[2], Emanuele Marsili[2], and Stephen Rolston[2] [1]PsiQuantum, 700 Hansen Way, Palo Alto, CA 94304, USA [2]Airbus Operations Ltd, Pegasus House Aerospace Avenue, Filton, Bristol, UK **Simulating non-trivial incompressible flows with a quantum lattice Boltzmann algorithm** David Jennings*, Kamil Korzekwa*†, Matteo Lostaglio*, and Paul Mannix*† *PsiQuantum, 700 Hansen Way, Palo Alto, CA 94304, USA* Richard Ashworth‡, Emanuele Marsili§, and Stephen Rolston¶ *Airbus Operations Ltd, Pegasus House Aerospace Avenue, Filton, Bristol, UK*	2025	End-to-end	Modest poly	Complete	Partially confirmed	Yes

This talk

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Quantum algorithms for CFD

- Origin of quantum speedup
- Pipeline for nonlinear differential equations
- Quantum lattice Boltzmann method

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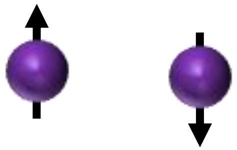
Origin of quantum speedup

Quantum data encoding

Single two-level system (qubit):



Its state:

$\Psi = \alpha_1$  $+ \alpha_2$ 

Probability amplitudes

$(|\alpha_1|^2 + |\alpha_2|^2 = 1)$

N qubits:

$\Psi = (\alpha_1, \dots, \alpha_{2^N})$

Encode exponentially many amplitudes

Quantum data processing

Unitary evolution of Ψ under the Schrodinger equation:

$$i\hbar \frac{d}{dt}\Psi = \mathrm{H}\,\Psi$$

Exponential quantum speedup for quantum simulations

Caveats

Evolution is linear → **Linear embedding**

Evolution is unitary → **Unitary encoding**

Readout problem → **Info extraction subroutine**

FRAMEWORK

Pipeline for nonlinear differential equations

Nonlinear differential equation

Linear embedding

Nonlinear ODE with n variables

$\downarrow$

Linear ODE with $\sim n^{N_C}$ variables

Linear differential equation

Discretization

$A\,\boldsymbol{x} = \boldsymbol{b}$

Linear system

Unitary encoding

$A \rightarrow U_A = \begin{pmatrix} A & * \\ * & * \end{pmatrix}$

Linear system encoded in a quantum circuit

Quantum linear solver (QLS)

Prepares a solution using q queries to U_A

Quantum encoded solution

Information extraction

Classical info

FRAMEWORK

Quantum lattice Boltzmann method

Incompressible Navier-Stokes equations (nonlinearity scales with Re)

Momentum equation $\partial_t \boldsymbol{u} + (\boldsymbol{u} \cdot \nabla)\boldsymbol{u} = -\nabla p + \nu \Delta \boldsymbol{u} + \boldsymbol{F}$

Mass conservation $\nabla \cdot \boldsymbol{u} = 0$

$\boldsymbol{u}$ – velocity field
p – pressure field
ν – kinematic viscosity
$\boldsymbol{F}$ – external force

Discrete-Boltzmann equation (nonlinearity scales with Ma)

$$\boxed{\partial_t f_m(\boldsymbol{r}, t) + \boldsymbol{e}_m \cdot \nabla f_m(\boldsymbol{r}, t) = \Omega_m(\boldsymbol{r}, t)}$$

$$p = c_s^2 \sum_{m=1}^{Q} f_m(\boldsymbol{r}, t)$$

$$\boldsymbol{u}(\boldsymbol{r}, t) = \frac{1}{\rho_0} \sum_{m=1}^{Q} f_m(\boldsymbol{r}, t) \cdot \boldsymbol{e}_{\boldsymbol{m}}$$

$\boldsymbol{e}_m \cdot \nabla f_m(\boldsymbol{r}, t)$ – streaming term

$\Omega_m(\boldsymbol{r}, t)$ – collision term

$f_m(\boldsymbol{r}, t)$ – discretized density of particles
$\boldsymbol{e}_m$ – unit velocity in direction $m \in \{1, \dots, Q\}$

$Q = 3/9/27$ – number of unit velocities available in 1D/2D/3D

FRAMEWORK

Quantum lattice Boltzmann method

Discretize space and time to obtain the lattice Boltzmann equation

$$\partial_t f_m(\boldsymbol{r}, t) + \boldsymbol{e}_m \cdot \nabla f_m(\boldsymbol{r}, t) = -\frac{1}{\tau}\left[f_m(\boldsymbol{r}, t) - f_m^{\mathrm{eq}}(\boldsymbol{r}, \mathrm{t})\right]$$

$$f_m(\boldsymbol{r} + \boldsymbol{e}_m \Delta t, t + \Delta t) = f_m(\boldsymbol{r}, t) - \frac{\Delta t}{\tau + \Delta t/2}\left[f_m(\boldsymbol{r}, t) - f_m^{\mathrm{eq}}(\boldsymbol{r}, \mathrm{t})\right]$$

Reformulate $f_m(\boldsymbol{r}, t)$ to only depend on t

$$f(t^*) = \sum_{\boldsymbol{r}^*} \sum_{m=1}^{Q} f_m(\boldsymbol{r}^*, t^*) \, |\boldsymbol{r}^*\rangle \otimes |m\rangle$$

$$\boldsymbol{r} = \boldsymbol{r}^* \Delta x$$

$$t = t^* \Delta t$$

Divide the integrated equation into streaming and collision distributions and apply Carleman

$$f^{\mathrm{C}}(t^*) = (1 + F_1) f(t^*) + F_2 f(t^*)^{\otimes 2}$$

$$f(t^* + 1) = S f^{\mathrm{C}}(t^*)$$

F_1, F_2 – linear and quadratic collision matrices

S – permutation matrix encoding streaming

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Performance of quantum LBE algorithm

- Central questions & benchmark problems
- Linear embedding accuracy
- Quantum speedup

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Main takeaways & areas for improvement

RESULTS

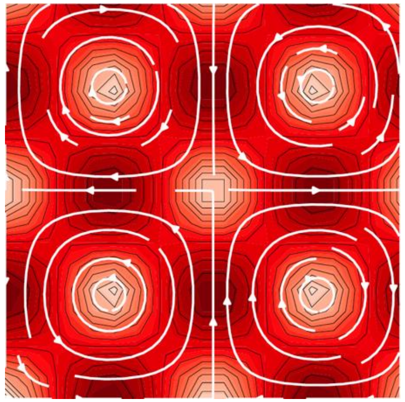
Central questions & benchmark problems

Central questions

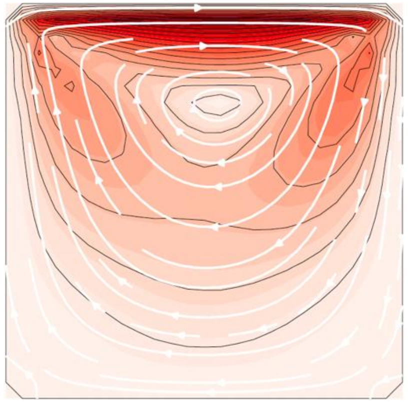
Linear embedding accuracy: how does the embedding error behave with growing N_C?

Quantum speedup: how does the quantum computational complexity grow with Re?

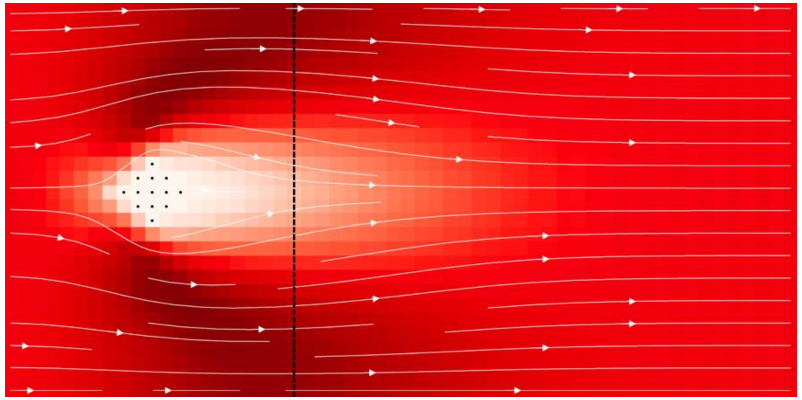
Benchmark problems



Driven Taylor-Green vortex



Lid-driven cavity flow



Flow past a cylinder

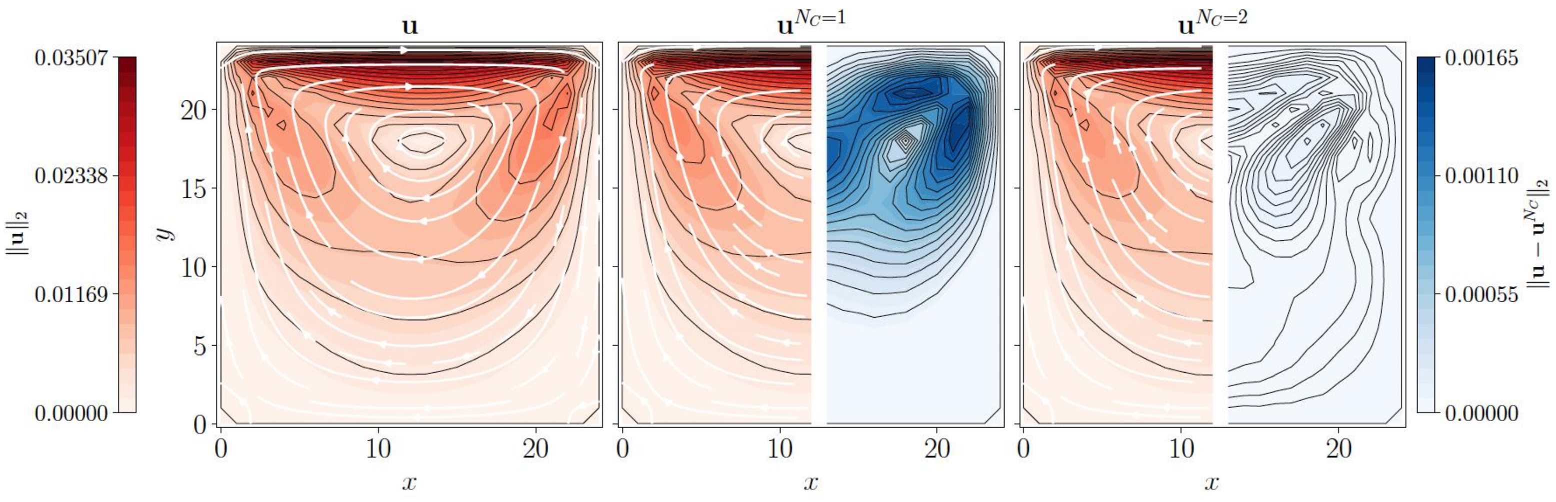
RESULTS

Linear embedding accuracy

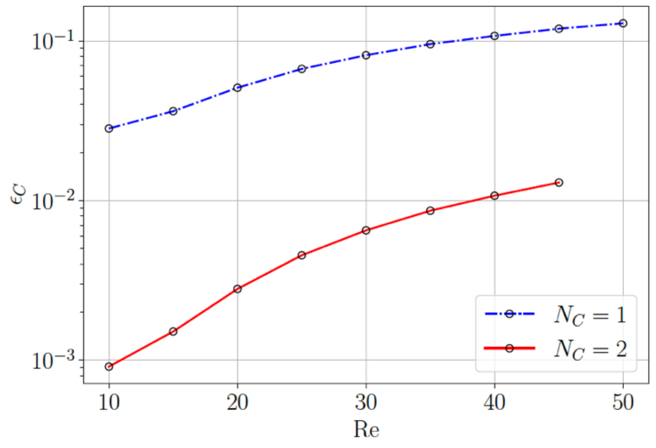
Embedding error measures

$$\epsilon_{rel}(\mathrm{t}) := \frac{\|\boldsymbol{u}^{\mathrm{N_C}}(\mathrm{t}) - \boldsymbol{u}(\mathrm{t})\|}{\|\boldsymbol{u}(\mathrm{t})\|} \qquad \epsilon_{\mathrm{C}}(t) := \max_{\mathrm{t}} \epsilon_{rel}(\mathrm{t})$$

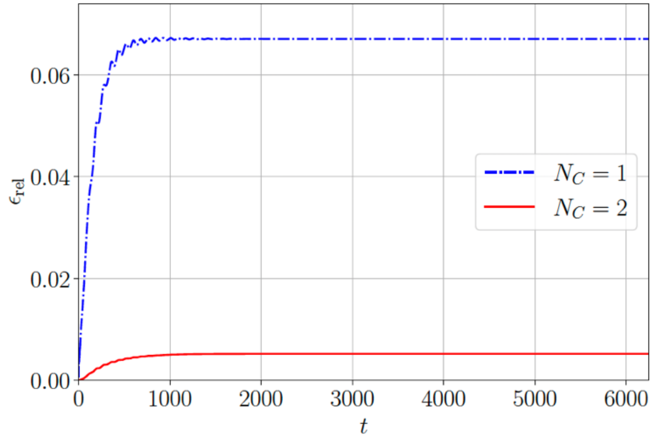
Steady state spatial embedding error for lid-driven cavity flow



Embedding error convergence



Embedding error boundedness



Quantum speedup

Quantum complexity contributions

- Initial state preparation
- Block-encoding $\mathrm{U_A}$ of the linear system A
- Solving the linear system A
- Information extraction

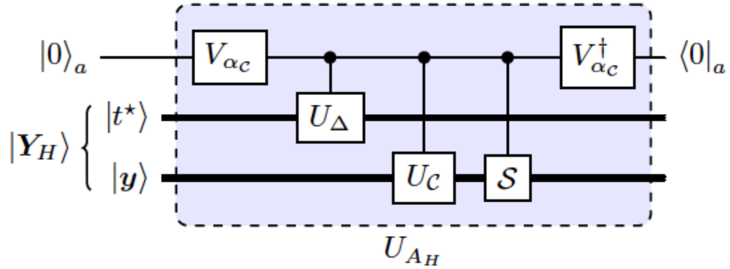
$$\mathrm{Complexity_Q} = g \times q \times q_M$$

Number of gates in a circuit $\mathrm{U_A}$

Number of circuits $\mathrm{U_A}$ used by QLS

Measurement info extraction cost

Compiled the core parts of the circuit to show that g brings just a constant overhead



Query complexity q is the main complexity factor

For drag force: $q_M = O\left(Re^{3/8}\right)$

For bulk observables expecting: $q_M = O(1)$

RESULTS

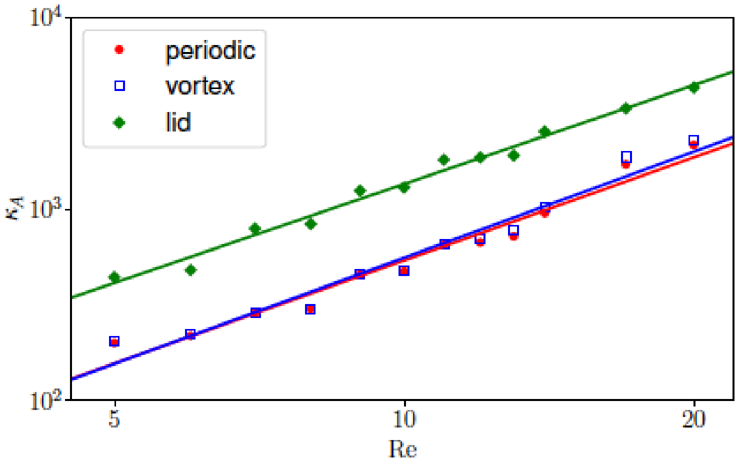
Quantum speedup

Complexity lower bound

$$\mathrm{Complexity}_Q \geq \mathrm{Re}^{\frac{3}{4}\left(1+\frac{D}{2}\right)} \qquad \text{vs} \qquad \mathrm{Complexity}_C = \mathrm{Re}^{\frac{3}{4}(1+D)}$$

At most lower-than-quadratic speedup

Numerical complexity estimate



For 2D problem with $\mathrm{N_C} = 2$:

$$\mathrm{Complexity}_Q \simeq \mathrm{Re}^{1.777} \times q_M$$

Modest quantum speedup

Bulk observables:	$\mathrm{Speedup} \simeq \mathrm{Re}^{0.473}$
Drag force:	$\mathrm{Speedup} \simeq \mathrm{Re}^{0.098}$

AGENDA

Main takeaways & areas for improvement

Carleman linear embedding method converges for benchmark CFD problems

Next steps:

- Code optimization
- GPU acceleration
- Analytical convergence proofs
- Alternative linear embeddings

Modest quantum speedup for CFD problems is in principle possible

Next steps:

- Preconditioning methods
- Novel quantum linear solvers
- More efficient observables

Thank you!

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